

- list
- tuple
- Set
- dect
- range

```
In [1]: l = []  
l
```

```
Out[1]: []
```

```
In [2]: len(l)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[2], line 1  
----> 1 len(l)  
  
TypeError: object of type 'int' has no len()
```

```
In [3]: l1 = [10,20,30]  
l1
```

```
Out[3]: [10, 20, 30]
```

```
In [4]: len(l1)
```

```
Out[4]: 3
```

```
In [6]: l1.append(40)  
l1.append(50)  
l1.append(60)  
l1.append(10)
```

```
In [7]: l1
```

```
Out[7]: [10, 20, 30, 40, 50, 60, 10, 40, 50, 60, 10]
```

```
In [8]: len(l1)
```

Out[8]: 11

```
In [9]: l1.append(50,20)
l1
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[9], line 1
----> 1 l1.append(50,20)
      2 l1

TypeError: list.append() takes exactly one argument (2 given)
```

```
In [10]: l1.append(['a','b','c'])
l1
```

Out[10]: [10, 20, 30, 40, 50, 60, 10, 40, 50, 60, 10, ['a', 'b', 'c']]

```
In [11]: l2 = [34,5.3,'prakash',1+2,True]
l2
```

Out[11]: [34, 5.3, 'prakash', 3, True]

```
In [13]: print(l)
print(l1)
print(l2)
```

```
[]
[10, 20, 30, 40, 50, 60, 10, 40, 50, 60, 10, ['a', 'b', 'c']]
[34, 5.3, 'prakash', 3, True]
```

```
In [15]: print(len(l))
print(len(l1))
print(len(l2))
```

```
0
12
5
```

```
In [16]: print(id(l))
print(id(l1))
print(id(l2))
```

```
2380799707392
2380805708608
2380815947264
```

```
In [17]: print(type(l))  
         print(type(l1))  
         print(type(l2))
```

```
<class 'list'>  
<class 'list'>  
<class 'list'>
```

```
In [18]: l1
```

```
Out[18]: [10, 20, 30, 40, 50, 60, 10, 40, 50, 60, 10, ['a', 'b', 'c']]
```

list membership

```
In [19]: 10 in l1
```

```
Out[19]: True
```

```
In [20]: 100 in l1
```

```
Out[20]: False
```

```
In [22]: ['a','b','c'] in l1
```

```
Out[22]: True
```

```
In [23]: l2
```

```
Out[23]: [34, 5.3, 'prakash', 3, True]
```

```
In [24]: l3 = l2.copy() # l3 here different memory is used but l2 data is copied from previous memory.  
         l3
```

```
Out[24]: [34, 5.3, 'prakash', 3, True]
```

```
In [25]: print(l1)  
         print(l2)  
         print(l3)
```

```
[10, 20, 30, 40, 50, 60, 10, 40, 50, 60, 10, ['a', 'b', 'c']]  
[34, 5.3, 'prakash', 3, True]  
[34, 5.3, 'prakash', 3, True]
```

```
In [26]: len(l3)
```

```
Out[26]: 5
```

```
In [28]: l3.clear()  
l3      # here l3 memory is cleared
```

```
Out[28]: []
```

```
In [30]: del l3  
l3
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[30], line 1  
----> 1 del l3  
      2 l3  
  
NameError: name 'l3' is not defined
```

```
In [32]: l2.count(5.3 # repeatation of number)
```

```
Out[32]: 1
```

```
In [33]: l1.count(10)
```

```
Out[33]: 3
```

```
In [34]: l1
```

```
Out[34]: [10, 20, 30, 40, 50, 60, 10, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [35]: l1[-1]
```

```
Out[35]: ['a', 'b', 'c']
```

```
In [36]: l1[100]
```

```
-----  
IndexError                                Traceback (most recent call last)  
Cell In[36], line 1  
----> 1 l1[100]  
  
IndexError: list index out of range
```

```
In [37]: l1[4:]
```

```
Out[37]: [50, 60, 10, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [38]: l1[3:-1]
```

```
Out[38]: [40, 50, 60, 10, 40, 50, 60, 10]
```

```
In [39]: l1[0:10:3]
```

```
Out[39]: [10, 40, 10, 60]
```

```
In [41]: l1[::1]
```

```
Out[41]: [10, 20, 30, 40, 50, 60, 10, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [42]: l1[::2]
```

```
Out[42]: [10, 30, 50, 10, 50, 10]
```

```
In [43]: l1[::3]
```

```
Out[43]: [10, 40, 10, 60]
```

```
In [44]: l1[::-1]
```

```
Out[44]: [['a', 'b', 'c'], 10, 60, 50, 40, 10, 60, 50, 40, 30, 20, 10]
```

```
In [45]: l1[::-2]
```

```
Out[45]: [['a', 'b', 'c'], 60, 40, 60, 40, 20]
```

```
In [46]: l1
```

```
Out[46]: [10, 20, 30, 40, 50, 60, 10, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [47]: l1[::-3]
```

```
Out[47]: [['a', 'b', 'c'], 50, 60, 30]
```

```
In [ ]:
```